Procedural Ragdoll Animation and Sprite Synthesis from Legacy 2D Game Art

A method developed in DERP using Doom's original monster sprites

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Author disclosure

I have a background in computer science and have followed computer graphics, game engines and physics simulation for much of my life. My understanding of their techniques and history comes from years of personal study and experimentation. I have never worked professionally in these fields.

The idea behind this project, its artistic direction and the decisions that shaped it are my own. AI coding agents assisted with much of the implementation and debugging.

These tools made it possible to explore and implement the ideas within a reasonable amount of time. This paper documents the resulting method, its limitations and the experiments that led to it. AI assistance was also used in preparing the paper.

Abstract

Legacy sprite animation describes a character through a small set of painted views and poses. Procedural motion requires configurations those images never show. The method described here derives an articulated relief from the original art, simulates its response to an impact, and renders the result back into a sprite. A front drawing defines the body, a side view constrains thickness, and rear artwork supplies its back colours. A compact skeleton generates the collapse and its contact with the environment.

Orthographic projection, discrete viewing directions, palette-based sampling and held poses keep the output close to the source game's presentation. The motion retains the character's facing, supports world-space flight and sliding, and publishes poses at selected impact and contact events. Once movement ends, the final image becomes the corpse sprite. The contribution is the integration of established graphics techniques into a procedural animation pipeline that uses the original artwork throughout.

1. The problem: generating poses from painted evidence

A sprite records a particular appearance. Its silhouette suggests the extent of the body, and its colours describe visible surfaces, clothing and carried objects. Depth and hidden anatomy remain ambiguous. Even a set of directional drawings may contain different limb positions, proportions or painted lighting. Many three-dimensional bodies could account for similar images.

The puppet is an inferred construction whose purpose is to provide enough volume and articulation for useful motion. The images constrain that construction, and simple assumptions fill the missing information. In Doom, the concrete problem is a death that responds to the killing blow while retaining the appearance and rhythm of the existing art.

Reusing painted death frames binds the animation to their sequence of poses. Timing changes stretch or compress that sequence. A skeleton supplies additional freedom: pelvis height, knee bend, torso angle and head motion can change in combinations absent from the drawings. An impact can produce a different order of collapse.

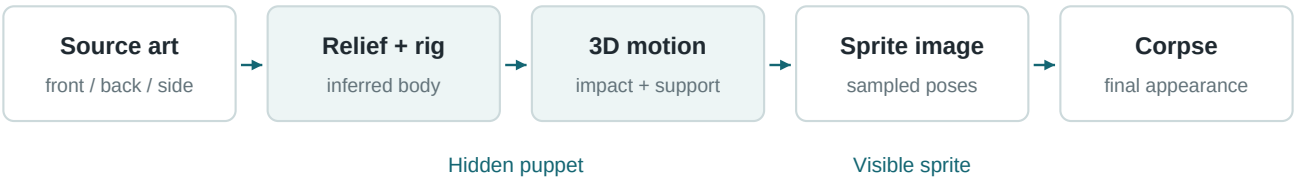


Figure 1. Original views define a hidden puppet. Its motion and contact produce poses that are sampled back into sprites.

The method separates body construction, motion and presentation. Geometry supplies articulation and self-occlusion at sprite scale. The simulation carries the impact through a body that can lose support, travel, hit an obstacle and settle. Presentation selects the images through which that motion becomes visible.

Continuity with the living monster matters at the first frame. The puppet keeps the direction the monster faced when it died. The hit is expressed in those anatomical axes, so a shot from behind remains a shot from behind. A rear drawing gives the generated body usable painted material on that side.

The puppet belongs to the presentation layer. Gameplay continues to use the original damage, state transitions and collision rules. The visual body inherits a bounded initial velocity from the game object and then follows its own trajectory. Its eventual position and silhouette can therefore differ from the gameplay corpse.

2. Inferring a deformable body

The puppet begins with an existing front-facing pain drawing. Its pose establishes the initial arrangement of the body, and its opaque regions define the relief's topology. A side drawing estimates depth. A rear drawing paints the back surface. These views retain distinct roles, avoiding a reconstruction that requires all directional paintings to describe exactly the same pose.

Silhouette as a depth prior

Silhouette inflation assumes that broad regions are thick and narrow regions are thin, a shape prior also used in sketch-based modeling such as Teddy [1]. Here, distances to the outline and local interior disks establish rounded cross-sections. Large disks describe the torso; smaller disks describe limbs and narrow objects. The estimator differs from Teddy's surface construction.

Broad measurements through the side view's legs, trunk and head constrain front and back reach. Aggregating rows reduces the influence of an extended arm or weapon. The two surfaces are closed around the outline, giving the drawing volume as it turns toward the ground.

Rear colours are mapped onto this front-derived mesh at corresponding heights and mirrored horizontal positions. A small search handles differences between silhouettes; missing samples fall back to front colours. The rear painting improves the visible material without supplying an independently reconstructed back shape. Overlapping objects, such as a rifle across the chest, can remain geometrically joined.

Articulation and painted material

Joint placement uses recognisable silhouette structure, especially legs that can be followed upward from the feet. Head detection supplies a neck and head region when the outline supports it. Hand-placed marks can supply joints for known drawings. These marks describe anatomy within the source image and require no replacement art.

Eleven simulated points span the head, torso and legs. Rigid parts joined by narrow blending regions drive the surface through linear blend skinning [2]. Each corner follows a weighted combination of part transformations while its faces retain their source-texel identity. Blending positions does not require blending the painted colours.

A carried weapon follows one rigid chest attachment. Its pixels retain the object's shape as the body bends. The compact puppet has no independent arm or foot articulation; arms merged into the torso follow that region.

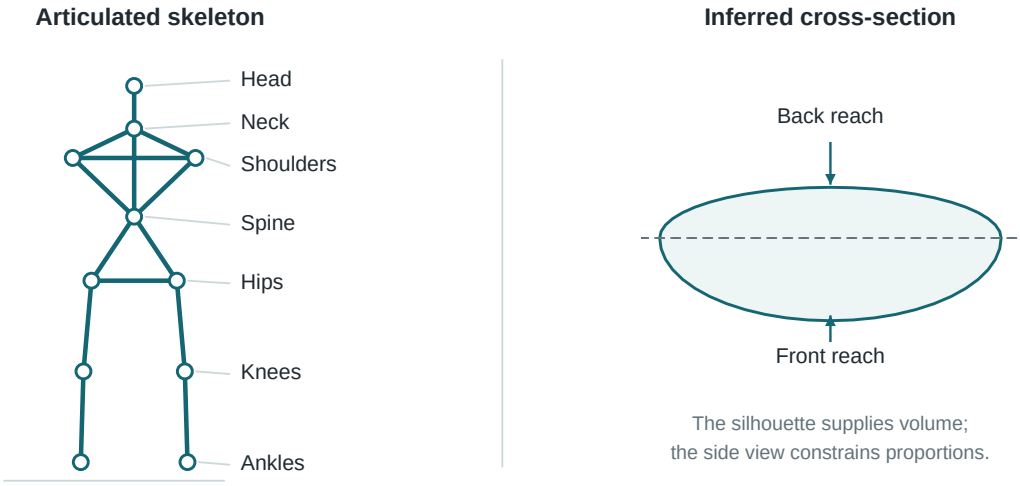


Figure 2. A compact skeleton supplies pose freedom. The relief supplies volume, with separate front and rear paintings on its surfaces.

3. Motion as the loss of support

A standing character remains upright because its joints resist collapse and its feet support its weight. Death can be described through the failure of that support. The impact supplies a disturbance; the release of joint resistance determines how it develops into a fall.

Hit direction, strength and position select affected regions and support schedules. A chest hit can recoil through the upper body while the legs remain firm. A weak low hit can buckle a leg while the trunk briefly stays upright. A stronger low hit can sweep the legs out, carrying the body over before it reaches the ground. Head and side hits disturb different parts, with their impulse strengths also responding to the measured blow.

Constraints, flight and internal damping

Following particle-based character physics [3], point motion uses damped Verlet integration. Distance constraints preserve bone lengths, additional links stiffen the torso, and temporary links resist changes in posture before weakening. Direct position correction places the solver in the position-based dynamics family [4]. Joint limits restrict folding and keep the knees near their hinge planes.

The order of support loss matters as much as the initial force. Early leg failure lets the pelvis descend before the chest turns. Retained leg support lets upper-body recoil develop first. A sweep briefly holds the trunk and hip structure while the unsupported legs move, then releases it as the body descends.

The visual body travels from its own world-space origin. Inherited game momentum supplies a bounded initial drift, with a small directional variation. A chest launch adds lift and rotation; a sweep supplies its own leg-driven turn. Internal reaction pushes are arranged separately so they do not erase the intended whole-body flight.

Damping distinguishes those two kinds of movement. A fitted rigid translation and rotation describe the body's overall motion. Residual point motion describes deformation relative to that fit. During flight, damping mainly reduces the residual, preserving travel and turn. This follows the rigid-motion-preserving damping principle discussed in [4]. Later, contact-sensitive damping suppresses motion near the ground. An explicit limit also restrains a tumble that would continue past a flat pose.

Directed variation

Seeded changes in push, yaw, knee disturbance and release timing produce different trajectories. Contact supplies further differences. A foot caught by a step can change which region reaches the floor first.

Steering remains an artistic control. It helps the body tip along the hit and favours a useful view of either painted side. It no longer resets the body to face the viewer. Strong leg sweeps retain their own imposed motion without this steering. The resulting dynamics combine physical constraints with staging choices made for the projected image.

4. Contact and guided dynamics

Contact changes the available support. A knee on a step, a torso over a ledge and a body against a wall impose different conditions on the same skeleton. Local floor sampling lets separate points rest at different elevations. Small collision volumes around the joints approximate floor tops, step faces and nearby walls.

Contact correction must avoid creating the motion it is meant to stop. A point pushed out of a wall receives the same positional correction in its current and previous states, preventing that repair from becoming a rebound. The wall-side test uses the immutable position at the start of the substep, because a joint correction can also move the previous state. Inward velocity is removed, wall friction reduces sideways skid, and upward contact motion is restricted.

Relative weights alter link corrections after wall contact. A stopped foot can then yield more readily to the trunk instead of dragging it around with equal influence. Open-floor motion retains the earlier equal-share correction. This is a targeted contact rule; the rigid-motion fit still uses equal points.

Extensions intended to loosen a lying pose are restricted to downward corrections. Otherwise, a body braced against a wall can satisfy them by climbing. Knee-plane limits, bounded folding and a tolerance near a straight knee reduce unstable or unreadable configurations.

A wall impact as a distinct event

A sufficiently fast torso impact can trigger a wall slam. The knees buckle, upper-body support weakens, and a short-lived pin restrains the torso while it descends toward sitting. Release depends on support and posture, with a finite time limit. These controls stage a readable collision while the same solver handles the resulting collapse.

Floor and wall effects follow the visual body's contact locations. Torso landing drives its floor burst and slap; a hard wall impact drives its wall response. A landed body that continues to slide can leave smears along that path. These effects do not feed forces back into gameplay.

Guidance through desired shape

The optional guided shot death uses a brief chest pull and soft targets suggested by the original death drawings. Rigid shape matching [5] fits each target to the body's current position and orientation before applying a partial correction. The targets describe internal shape while the current trajectory and contact determine placement.

The pull weakens near landing and eventually releases. Its pose keys are authored guidance, while the articulated solver supplies the transitions and environmental response. Free, swept and guided falls share the same relief and sprite-synthesis stage.

5. Sprite synthesis as an appearance constraint

Displaying the inferred relief directly exposes its depth estimate, limited anatomy and newly revealed surfaces. Rendering it through a virtual camera produces a new image with a controlled silhouette and resolution. The generated card is an image-based impostor [6] whose purpose also includes preserving the source game's appearance.

A camera that follows the view

Orthographic projection keeps a surface element's image scale independent of depth within the puppet. Horizontal viewing direction follows the player through the fall in discrete steps. Hysteresis around direction boundaries prevents small view changes from repeatedly switching between adjacent images.

The camera does not orbit to improve a difficult corpse pose. An earlier side-on adjustment could look like the body rolling after landing. The current view follows the observer, leaving the simulation responsible for the depicted motion.

Downward inclination depends on both the body and the actual viewing elevation. An upright body receives little downward tilt. A lying body receives an elevated view only as far as the observer's angle allows, up to a modest cap. This limits the far half of the projected body appearing to hover when its image is placed on an upright card.

Sampling the painted surfaces

Rasterization operates at the source art's scale. Visible samples retain discrete palette colours, with palette-based darkening for exposed thickness. Projection can hide, duplicate or omit texels; it preserves the source material's origin and character rather than a one-to-one pixel arrangement.

Front and rear paintings remain attached to their respective relief surfaces. Faces that turn away during folding are also available as darker, depth-biased coverage. This repair closes visual gaps caused by the economical surface representation. It adds no new anatomical information.

Source-coordinate wounds follow the same moving regions. The completed picture receives lighting normals inferred from its silhouette. The card's world location selects its surrounding sector lighting and liquid treatment, so a thrown body can enter a differently lit region.

The image in the world

The result is an upright, viewer-facing billboard. Its internal depth buffer resolves self-occlusion during synthesis; the world sees the ordinary sprite plane. A small downward placement offset helps the lowest rows meet the floor as the body lies down. Sprite-plane clipping remains part of this representation.

Shadows now use the current synthesized silhouette and its tight opaque bounds. They can therefore follow the fall and final corpse instead of projecting the original standing drawing. Contact and selected-light floor shadows use the image; eligible flashlight wall silhouettes use it too. These are image-based shadows, with the corresponding approximation of the hidden body.

6. Temporal sampling and the end of motion

A stable trajectory and a readable sprite animation have different timing requirements. The solver needs small steps for constraints and contact. The visible animation needs held poses, with enough samples to communicate the hit and landing.

The method separates the simulated trajectory from the displayed pose sequence. Early poses change more quickly to expose recoil. Later airborne poses have a maximum hold duration, while grounded motion can follow the monster's original death-frame pace. The visual body's horizontal centre can interpolate between tics while the articulated shape remains held.

Sampling meaningful events

Uniform samples can miss a short event. A recoil peak or impact may occur between two selected configurations, leaving one image airborne and the next already settled. Increasing force alone does not guarantee that the missing action becomes visible.

Additional poses are published at the top of a throw, initial landing, torso descent, a wall slam and final rest. Regular holds retain the cadence; event samples preserve mechanically important moments. These tests are deliberately limited to identifiable events in the body simulation.

Letting support determine the endpoint

The first detected landing and a completed collapse are different states. The head may approach the floor while the torso still has a long way to fall. The settling deadline therefore starts from a torso-down condition. A brief grace period also delays stronger grounded damping.

Quiet motion can end the fall early. A later deadline requires low horizontal travel, and a further limit eventually ends a slowly moving body. An overall lifetime cap bounds exceptional cases. The quiet metric sums stored point displacement, so these tests describe motion rather than an anatomical correctness score.

From an evolving body to a persistent image

At rest, the current picture becomes the corpse sprite. Keeping that image preserves the proportions, material placement and pose seen at the end of the fall. For the final corpse, the same picture is reused from every viewing direction.

The billboard still responds to ordinary world lighting. A recorded supporting floor can carry it vertically when a lift moves. Its internal image and view remain fixed, and the simulation and repeated synthesis have ended. This supplies a stable appearance without a hand-off to a separately painted corpse pose.

7. Evaluation, assumptions and scope

Development first established that a skeleton could produce useful poses absent from the painted death sequence. Thickness made those poses legible when lying down. An eight-view visual hull was also tested: inconsistent strides damaged the geometry, and mixed painted lighting produced mottled colours. The current relief keeps front-derived geometry, uses the side view for depth proportions, and applies front and rear colour to coherent surfaces.

Artist-built voxel models served as measurement references during the depth experiments. More exact depth still could not reliably separate objects overlapping in the front drawing. This supported the simpler relief for the intended image resolution. It does not establish that less geometry is preferable for every character or viewing condition.

Controlled engine scenarios

A dedicated arena now provides repeatable settings for flat floors, stairs, drops, walls and clutter. Different initial headings expose the transition from the living sprite. Controlled hit locations and impulse levels isolate response families, and alternative observer positions reveal projection effects.

The fixture sends real hitscan attacks through the engine, with fixed aligned pellet counts and damage chosen to produce a kill. These are controlled inputs, without the normal weapon spread. A stable test seed supports repeated comparisons. Motion logs record contact, travel, rotation and the reason a body stops; image sequences retain the actual synthesized result.

This is stronger diagnostic evidence for the current implementation than the earlier Python proxy, which differs in motion, environment and sampling. Short contact sheets can still end before the complete settling lifetime. The apparatus enables comparisons; its presence is not a completed statistical or perceptual evaluation. This revision was checked against source and existing local review material, without a new engine benchmark.

Remaining assumptions

Joint discovery needs recognisable silhouette structure. The compact skeleton restricts arm and foot motion, and front-derived relief cannot recover all hidden anatomy. Rear artwork improves colour coverage without resolving those geometric ambiguities. Point-sphere contact leaves gaps between the collision model and the visible surface; self and corpse-to-corpse collisions remain outside the model.

The fixed corpse image gives up later view consistency. Save loading does not reconstruct its transient rendering state. Broader anatomical coverage, sustained performance with many bodies and comparisons with other synthesis methods require separate evaluation.

References

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[3] Thomas Jakobsen. 2003. **Advanced Character Physics.** Game Developer, 21 January 2003. Publisher's article. This citation identifies the online publication; the work was originally presented at GDC 2001.

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[5] Matthias Müller, Bruno Heidelberger, Matthias Teschner, and Markus Gross. 2005. **Meshless Deformations Based on Shape Matching.** ACM Transactions on Graphics 24(3), 471-478. doi:10.1145/1073204.1073216. Author's paper.

[6] Paulo W. C. Maciel and Peter Shirley. 1995. **Visual Navigation of Large Environments Using Textured Clusters.** Proceedings of the 1995 Symposium on Interactive 3D Graphics, 95-102. doi:10.1145/199404.199420.

These references establish the relevant method families. The relief uses a different depth estimator from Teddy. Skinning follows the weighted skeletal-transform formulation discussed in [2]; pose-space corrective deformation is outside this method. Guided motion uses rigid fitting from shape matching within an articulated solver. The final card is an impostor, without the scene-wide hierarchical cluster selection of [6].

Appendix A. Reproduction parameters

A.1. Source images, relief and skinning

This records the live working tree inspected on 20 October 2026: local savepoint `2f573f7691ec96c6de694d2f2a3ced5389d1a091` plus subsequent uncommitted motion and shadow changes. It is a parameter record, without a claim of optimality. Distances use Doom map units. Local x points right in the front drawing, y away from its original viewer, and z up. Let w and h be image dimensions and T the patch top offset; T can differ from h.

Quantity	Specification
Source views	Pain-state front rotation defines geometry and front/wall colours. Doom rotation 3 supplies depth proportions; rotation 5 supplies rear colours. Respect the views' flip flags.
Rear correspondence	With patch left offsets Lf, Lb and top offsets Tf, Tb: rb = r + Tb - Tf. Unflipped rear: cb = Lb + Lf - c - 1; flipped rear: cb = c - Lf + Lb.
Rear fallback	Search dc = -3..3, dr = -2..2 for an opaque sample minimizing dc squared + 2dr squared. Ascending dr/dc order breaks ties. Fall back to the front texel if no sample exists, or to front colours if no distinct rear image exists.
Rig selection	Exact stock rig by lump name, dimensions and FNV-1a pixel hash; otherwise automatic leg discovery. Head detection is optional.
Automatic leg proportions	For tracked length L, floor F and crotch C = F + L: hip = C + 0.35L, ankle = F + 0.22L, knee halfway. Require six rows and L within 20-60% of the supported body height.
Rest skeleton	Eleven points: head, neck, shoulders L/R, spine, hips L/R, knees L/R, ankles L/R. All begin at local y = 0.
Skin parts	Pelvis, chest, head, thigh/shin L/R and rigid carried weapon. Arms follow torso regions.
Tube estimator	Euclidean distance d to transparency with transparent padding. Local interior disks propagate their largest covering radius R. Set e = d - 0.5, r = R - 0.5; H = sqrt(max(0, e(2r - e))).
Smoothing	Opaque-only Gaussian blur: horizontal sigma = 1.2; vertical sigma = clamp(0.04h, 1.5, 4). Support radius ceil(2.5 sigma); renormalize opaque samples.
Side bands	[0, 0.45T), [0.45T, 0.8T), [0.8T, 1.2T), using row centres. Runs must be at least two pixels wide.
Side reaches	Sorted index floor(0.4(n - 1)) separately for front/back. Leg rows use half the upper-median run width. For the trunk/head run [a, b) nearest the origin, use max(0.5, -a), max(0.5, b).
Depth adjustment	Interpolate at 0.225T, 0.625T, 0.9T. Clamp requested reach to [0.3P, 1.3P] for row tube peak P, then scale H by reach/P. Missing side data retains symmetric depth.
Torso construction	Neck from detector, fallback 0.8T. Spine at hip + 0.45(neck - hip), head at T - 4. Shoulders three units below neck. Half-span max(2, 0.8 min(m - left extent, right extent - m)), fallback 7, about hip-centre m.
Skin transitions	Hip and left/right leg blends span three units; knee, chest and neck blends span two. Retain and normalize the four largest influences. Weapon weight is 1 on its carried pixels and follows the chest frame.
Mesh	Front/back quad per opaque texel; walls at opaque/empty four-neighbour boundaries. Corner depth averages four neighbours, counting empty as zero.
Face appearance	Front/back dim base 1, wall base 0.7; multiply by 0.4 + 0.6 abs(n_y) of the rest-surface normal. Back faces select rear colour, other faces front colour.

For weights, Q(z, j, a) = clamp((z - j)/(2a) + 0.5, 0, 1). Set u = Q(z, hip, 1.5), c = Q(z, chest, 1), n = Q(z, neck, 1). Pelvis/chest/head weights are u(1 - c), uc(1 - n), ucn. Left share is clamp(0.5 - (x - split)/3, 0, 1); right is its complement. With side share s, thigh/shin weights are (1 - u)sQ(z, knee, 1) and (1 - u)s(1 - Q(z, knee, 1)). A corner is the sum of wj(Rj p + dj), with Rj = Bnow Brest-transpose and dj = onow - Rj orest.

A.2. Integration, constraints and damping

Quantity	Specification
Time	35 tics/s. Four substeps per tic (140/s), dt = 0.25 tic. Six ordered constraint passes per substep.
Gravity	(0, 0, -0.45) units/tic squared; add gravity times dt squared per substep.
Stored motion	v = p - p_prev. Injecting velocity u in units/tic subtracts u/4 from p_prev.
Structural links	Fourteen hard distance links k = 1; six permanent soft cross-body links k = 0.15; muscle links use their current coefficient each pass.
Correction shares	Until wall contact, endpoint a receives 0.5. After first wall contact, share = (1/wa)/(1/wa + 1/wb). The other endpoint receives the complement.
Relative weights	Head 0.07, neck 0.08, shoulders 0.10 each, spine 0.15, hips 0.10 each, knees 0.10 each, ankles 0.05 each. Used by link/linkDown after wall contact, not by the rigid fit.
Torso limit	Neck-to-each-hip distance at least 90% of rest, k = 0.5.
Hip direction	Thigh dot limits -0.5 against pelvis front and 0.34 against pelvis up. Rotational correction moves current and previous leg positions together.
Knee direction	Wrong-way sign test < -0.15; reflect current/previous knee about hip-ankle axis. Lateral shin deviation from the hinge plane is limited to 0.35 shin length.
Leg reach	Hip-ankle minimum 55% of thigh + shin length, k = 0.5. Grounded extension targets 85%, k = 0.03, using downward-only corrections.
Hip relaxation	While lying, neck-knee target 95% of its standing distance, k = 0.03, only when knee clearance < radius + 4. Downward-only correction avoids lifting a braced body.
Rigid fit	Equal-point centre and average velocity; solve the inertia system for angular velocity. Residual = point velocity minus fitted translation and rotational velocity.
Flight damping	Retain rigid translation/rotation; multiply internal residual by 0.985 per substep. Additional anti-overturn control retains 0.7 of the relevant horizontal spin when trunk-axis z < 0.1 and spin continues past flat.
Grounded damping	After three tics from torso-down, use per-point global damping: 0.9 if clearance < sphere radius + 1.5, otherwise 0.985.
Lying/down	Lying requires sticky landed plus spine clearance < 8 or neck < 7. Down also accepts a landed, seated wall-slam body. First down records lyingAt.
Travel bound	Once horizontal local mean distance reaches 136 units, remove outward mean velocity so it stays near the sampled contact region.

A substep computes stored displacement, applies the selected damping and any active wall pin, and saves the immutable starting pose. It then predicts new positions, applies optional shape guidance, and performs six passes of bone links, soft links, muscles, limits and point contact. Friction follows once per substep. Contact corrections may themselves alter stored motion as specified in A.5.

After four substeps, constrain travel, apply steering unless excluded by sweep/slam, update contact events and increment the tic. The motion metric is E = sum over eleven points of squared length(p - p_prev), in map-units squared. It includes whole-body stored motion; it is unweighted and is not kinetic energy or divided by dt squared.

Sources: `ragdoll.cpp` (`Ragdoll::tic`, `shareOf`, `link`, `linkDown`, `limits`, `rigid`, `damp`, `keepNear`, `ragdollView`).

A.3. Impact and world-space travel

Hitscan impacts in one tic are averaged per target. Normalized height is clamp((hitZ/max(16, object height) - 0.45)/0.3, -1, 1), fallback 0.2. Lateral offset is measured across the shot in target radii. The hit is then converted from shot coordinates into the monster's retained, possibly mirrored anatomical axes. Push speed s is the gameplay momentum change beyond expected friction.

Selection priority: head hit; low hit below -0.2 with s >= 2 selects sweep; otherwise low hit or s < 1 selects legs; otherwise absolute lateral offset > 0.5 selects side; remaining hits select chest. Head requires height >= drawn neck and horizontal distance from head centre <= half-width + 1. Let F = clamp(s/6, 0.6, 1.4), B = clamp(s/3.5, 0.8, 1.6), (tx, ty) be body-space shot direction and a the hit-side sign.

Family	Base internal velocity; affected points	Legs	Trunk	Neck
Chest	2.2F(tx, ty, 0) + (0, 0, 0.4); spine, shoulders, neck	9 / 6	5 / 10	2 / 6
Legs	B(1.1a, 0.5ty, 0); struck hip/knee, spine	0 / 5	10 / 10	6 / 10
Side	B(1.6a, 0.7ty, 0); struck shoulder, neck, spine	4 / 8	5 / 10	0 / 4
Head	1.2B(tx, ty, 0); head	0 / 2	0 / 2	0 / 2

Muscle entries are hold/fade tics. Add independent [-2, 2) jitter to each hold and [0, 6) to the leg group. Ordinary impulses retain the earlier [0.8, 1.2) per-point scale, shared x drift [-0.4, 0.4), yaw coefficient [-0.1, 0.1), forward knee kicks [0.2, 1.1) and outward kicks [0, 0.4). Remove their mean horizontal drift. Additional hit-directed tip is 0.15B for chest, 0.3B for the other ordinary families.

Motion control	Specification
Inherited travel	Add gameplay momentum once, magnitude capped at 4.5 units/tic; turn its direction by a seeded signed angle of 2-7 degrees. The visual origin and contact then evolve independently.
Chest launch	h = clamp((s - 0.8)/2.7, 0, 1.5). At tic 0 add min(h, 1)(tx + 2q, ty, 0) + (0, 0, 2min(h, 1.2)), q uniform [-0.25, 0.25). Add centred somersault coefficient 0.05min(h, 1).
Strong low sweep	l = clamp((-high - 0.2)/0.6, 0, 1); S = clamp(s/3.5, 0.7, 1.4)(0.6 + 0.4l). Reverse somersault -0.2S radians/tic, lift 1.0S units/tic; one ankle gets 0.8S along the shot.
Sweep support	Legs/neck muscles zero. Trunk and extra spine-knee links remain at 1 until spine clearance < 20, then fade over three tics. Initial yaw coefficient is signed(0.52-0.79)/10 radians/tic; subsequent dynamics determine the turn. Strip internal drift, then add inherited travel.
Steering	Ordinary falls tip along the hit and choose the nearer painted-side orientation among +/-up and +/-toward-eye. Twist target clamped to +/-0.15 radians/tic and approached halfway. No steering for sweep or after a wall slam.

Sources: `ragdoll.cpp` (`startRagdoll`, `launch`, `tip`, `steer`, `noDrift`).

A.4. Optional pose guidance

The guided branch uses the inherited travel and chest launch in A.3. Its additional pushes act inside the moving body. The following settings describe its preferred shape progression and release.

Control	Specification
Guided eligibility	Compatible rig, chest family, anatomical ty > 0.7, chance < 0.67 and guidance enabled. Currently the zombieman pain rig is compatible.
Ratchet/support	At tics 0-2, internal chest pull = 0.9 clamp(s/4, 0.7, 1.8)(tx + q, ty, 0.15), q uniform [-0.25, 0.25); head receives 60%. Knees kick at tic 4. Internal pushes preserve inherited travel and launch.
Guided support	Knee kicks [0.2, 0.8) at tic 4; leg hold 4 + U[0, 2), fade 2. Trunk coefficient 0.8 fades over three tics after hip contact; neck 0.5 fades over one after chest contact.
Guide strength	Base 0.35, ramp 0 through tic 2, 1/3 at 3, 2/3 at 4, 1 from 5. Multiply by clamp((spine clearance - 10)/8, 0, 1) and clamp(1 - (t - 17)/6, 0, 1). Zero after landed or during wall pin.

Key tic	Spine arch (deg)	Head (deg)	Hip (deg)	Knee (deg)
0	0	0	0	0
2	15	30	0	5
7	35	55	15	25
12	25	50	60	80
17	10	40	75	95

Keys use smoothstep interpolation. Per-substep shape strength is 1 - (1 - min(k, 0.99))^(1/4), retaining 5% of correction as stored displacement. Grounded points accept vertical pulls only. Sources: `ragdoll.cpp` (`startRagdoll`, `launch`, `pushInside`, `steer`, `guideBefore`, `keyAt`, `shapeMatch`).

A.5. Contact, wall slams and retirement

Let clearance be the point's height above its sampled floor. Contact remains joint-sphere based; there are no full limb capsules, mesh contacts, self collisions or inter-corpse collisions.

Quantity	Specification
Point radii	Head 4; neck/shoulders 2.5; spine 4; hips 3; knees 2; ankles 1.5.
Floor sampling	Four-unit cells, 80 x 80 grid spanning 320 x 320 units around kill. Cache cell-centre sector identities; query live floor heights while active. Clamp off-grid queries to edge cells.
Across solid walls	Grid samples reached across a cached full wall use the kill sector to avoid false drop-offs through wall boundaries.
Floor correction	Up to three deepest-cell corrections per point/pass. Rounded drop edges radius 2 when neighbour is lower by > 0.5. If embedded more deeply than radius + 6, prefer lateral exit through an available lower edge.
Contact velocity	Shift current and previous positions together for sideways repairs, then remove inward velocity. Floor tops remove downward motion.
Floor friction	D = accumulated upward correction in a substep, s = horizontal stored displacement. If s <= 0.6D, cancel horizontal displacement; otherwise keep fraction 1 - 0.5D/s.
Wall sampling	Lines overlapping the kill's +/-160-unit square. Skip two-sided lines with opening >= 16 and step above kill floor <= 16. Other lines constrain below their cached top; closed/narrow openings and one-sided lines are full barriers.
Wall-side test	Preserve the side occupied by the immutable substep-start point, including after joint corrections have moved p_prev. Position repair adds no displacement velocity.
Wall friction	Remove inward and upward motion. Lose 50% of horizontal tangential skid per substep.
Slam event	First hard tracked upper-body wall contact (head, neck, shoulders or spine) while not landed: impact speed > 1.5 units/tic, spine clearance > 14, trunk-axis z > 0.4. Knees kick 1.2 away from wall and -0.3 vertically; hips kick 0.36 away.
Slam pin	At most 40 tics, while spine > 10 and hips not seated. Legs release, trunk strength capped at 0.5, shape guidance zero. Torso pin cancels outward/ascent and halves lateral velocity.
Pin descent	Maximum 0.1 units/tic for first three tics, then min(3, 0.1 + 0.4(age - 3)). Release when both hip sphere surfaces lie within 1.5 of the floor.
Landed flag	Sticky from spine clearance < 6, neck < 5, head < 4.5, or seated after slam. Detection begins after the third simulated physics tic.
Torso-down clock	Record lyingAt on the lying/down condition in A.2, separately from first contact. Three-tic grace before grounded damping.
Quiet sleep	Landed and E < 0.002 for eight consecutive tics, E defined in A.2. Reset counter on failure.
Other sleep tests	Down for at least 40 tics and horizontal mean displacement this tic < 0.2; or down for 60 tics; or total age 175 tics.
Contact effects	Landing burst at torso-down, wall response at the first sufficiently hard tracked upper-body wall contact, sliding smears along the visual path. Older actor-based effects skip active ragdolls.
Supporting floor	On sleep, retain the sector under the spine and its floor height. Apply later floor-height differences to the baked card.

The clocked limits bound motion but do not score anatomical correctness. The torso-down deadline avoids starting a collapse timer from an early head contact. Floor heights are live during the fall; wall tops are cached.

Sources: `ragdoll.cpp` (`floorsNear`, `ground`, `rub`, `struck`, `pin`, `seated`, `steer`, `ragdollView`, `dollPlace`); `spray.cpp` (ragdoll exclusions).

A.6. Projection, timing and shadows

Quantity	Specification
Projection	Orthographic, one image pixel per map unit, aimed at the posed 3D bounding-box centre.
View direction	Eight directions at 45-degree intervals. Retain last direction while wrapped difference from it is < 32.5 degrees (22.5 + 10). No automatic orbit.
Inclination	L = clamp(1 - upright.z, 0, 1). Let looks be actual viewer look-down angle to the body's horizontal midpoint, using the placement floor reference. Pitch = 5 degrees times round(clamp(looks, 0, 30)L/5).
Canvas	D = maximum front + back depth. S0 = ceil(sqrt(w squared + h squared + D squared)) + 8; round upward to a multiple of four.
Surface sampling	Two triangles per quad, pixel-centre coverage, nearest biased depth. Retain both face orientations.
Away-facing coverage	A = 1 for a face turned away from the camera, otherwise 0. Depth = projected depth + A. COLORMAP level = clamp(round(24(1 - face dim)) + 6A, 0, 31).
Wounds	Source mask 1 omits all faces for that texel. Rim values >= 2 replace front-face colours before darkening.
Sprite normals	Infer from completed silhouette; depth scale 1.75, vertical gradient scale 0.6. RGBA8 holds palette index, coverage, normal x/y. No puppet depth or mesh-normal image is exported into world rendering.
Card placement	Upright, viewer-facing plane. Track visual horizontal mean independently of held shape. Lowest visible row follows the lowest sphere surface, with additional downward offset 3L units. Apply liquid sinking separately once landed.
Pose holds	Every two tics through tic 8; subsequently airborne every min(monster cadence, 5); after the initial landed event use first death-state duration clamped 2-10 tics.
Event poses	Tic 1, throw apex, initial landed event, torso-down, wall slam and final rest. Apex: previous mean rise > 0.3 and current mean rise <= 0 while airborne.
Image updates	Shown pose, camera angle step, pitch step, mirror or wound signature change. Cache posed corners until shown pose changes.
Opaque bounds	Track left/top inclusive and right/bottom exclusive bounds of generated coverage; use them for shadow dimensions and sampling rather than the padded square.
Floor shadows	Current silhouette. Width = opaque width times min(1.7, 1 + height/160); base length = 0.65 width; opacity = 0.28 sectorLighting/(1 + height/48), omitted below 0.015.
Selected-light shadows	Current card and opaque height; projected floor length capped at 192. Flashlight wall projection uses a 6 x 8 grid, ordinary solid/shootable eligibility and nearest-six caster selection.
Bake	Keep one final image at its last direction and pitch; the same picture is reused from every viewing direction. Supporting-floor motion and world lighting can still affect placement/appearance.

Shadows project the generated image, rather than making a fresh light-view rendering of the hidden relief. Billboard depth and its ordinary clipping convention are retained. Sources: `ragdoll.cpp` (`ragdollCard`, `dollPlace`, `ragdollWhere`); `geometry.cpp` (card emission, lighting and shadow dispatch); `shadows.cpp` (`addEnemyShadow`, `flashlightSilhouette`).

A.7. Test protocol and reproduction boundary

The arena generator supplies six map cases: flat floor; descending 16-unit stairs; a 96-unit drop; a wall 48 units behind; clutter; and close walls 22 units behind. Each places a row of zombiemen, shotgun guys or imps in four initial headings. Captures can retain one victim, change kill tic, or move the observer across the flight.

The engine fixture fires one, three or seven aligned attacks for pistol-, shotgun- and super-shotgun-style inputs. Each attack has fixed damage 10, and victim health is set so the last pellet kills. It uses the engine's real hitscan and thrust path but omits normal weapon spread. Chest aim is 55% of object height, head height minus 5, legs 20%, and lateral offsets +/-0.6 radius.

Record	Purpose
Source snapshot	Retain the exact source tree as well as its savepoint identifier; A.1 includes live changes beyond that commit.
Character inputs	Original WAD/palette, source front/side/rear patches, rig metadata and carried-pixel mask.
Hit and initial state	Victim type, position/facing, incoming momentum, hit measurements, chosen response and initial random seed.
Environment/view	Arena map, moving floor state, viewer path and capture schedule.
Per-tic diagnostics	Travel, velocity, fitted turn and shoulder twist, clearances, contacts, slam state and quiet/deadline/late/cap stopping reason.
Images	Keep full-resolution event poses and the actual final baked image; a small contact sheet can conceal silhouette details.

Normal mode seeds cosmetic variation from object address and level tic. Test mode substitutes spawn location/type for the address-dependent identity, making controlled local repeats more stable. Same-tic mirroring still depends on observation order. Neither mode establishes cross-platform bitwise equivalence for floating-point constraints.

Simulation advances when its view is requested and catches up to the current game tic. The visual body's world trajectory is integrated internally, while floor queries can observe live heights at access time. Exact comparisons therefore need the environment and observation schedule as well as a seed. Save loading does not reconstruct the transient doll or its baked texture.

The default capture uses twenty frames three tics apart, about sixty tics. It can miss later retirement paths; tests of the full 175-tic limit need longer capture or logs. Existing review images and research notes are qualitative development records. Earlier Python-proxy counts and source-comment comparisons are not presented here as current engine failure rates. No new engine or perceptual benchmark was run for this revision.

Sources: `tests/ragdoll_arena.py`; `platform/scene3d/ragdoll_test.cpp`; `ragdoll.cpp` (`startRagdoll`, `ragdollView`); `scene.cpp` (frame capture). Geometry definitions are in `ragdoll_body.cpp`, `sprite_normals.cpp`, `death_joints.h` and `death_rigs_stock.h`.